DAY 5 – IMPORTING AND CLEANING DATA IN R

1. Introduction to Data Importing

In real-world Data Analysis, data is generally available in different file formats:

1. CSV files
2. Excel files
3. Text files
4. Database tables

Before analyzing data, we need to import the data into R.

Example: A company provides an employee salary report in Excel. We need to import that file into R and clean it before performing analysis.

2. Import CSV File

CSV stands for Comma-Separated Values.

Each row represents a record, and commas separate the values.

Syntax:

```
data <- read.csv("filename.csv")
```

Example:

```
employees <- read.csv("employees.csv")
```

Display first 6 records:

```
head(employees)
```

Display first 10 records:

```
head(employees, 10)
```

Display last records:

```
tail(employees)
```

3. Import Excel File

To import Excel files, we use the `readxl` package.

Step 1: Install package

```
install.packages("readxl")
```

Step 2: Load package

```
library(readxl)
```

Step 3: Import Excel file

```
employees <- read_excel("employees.xlsx")
```

Import a specific sheet:

```
employees <- read_excel(
  "employees.xlsx",
  sheet = "Sheet1"
)
```

4. Explore Imported Data

After importing data, we should understand its structure.

Function	Purpose
`head()`	First 6 records
`tail()`	Last 6 records
`str()`	Data structure
`summary()`	Statistical summary
`dim()`	Rows and columns
`nrow()`	Number of rows

`ncol()`	Number of columns
`names()`	Column names

Example:

```
head(employees)

str(employees)

summary(employees)

dim(employees)

nrow(employees)

ncol(employees)

names(employees)
```

5. Understanding Missing Values (NA)

Missing values are represented by `NA` in R.

Example:

```
employees <- data.frame(
  Name = c("Amit", "Rahul", "Priya", "Neha", "Vikas"),
  Department = c("Sales", "IT", "HR", "Sales", "IT"),
  Salary = c(30000, 45000, NA, 55000, 60000)
)
```

Check missing values:

```
is.na(employees)
```

Count missing values:

```
sum(is.na(employees))
```

Column-wise missing values:

```
colSums(is.na(employees))
```

6. Handling Missing Values

There are different ways to handle missing values.

Method 1: Remove rows containing missing values

```
clean_data <- na.omit(employees)
```

Method 2: Replace missing salary with average salary

```
employees$Salary[
  is.na(employees$Salary)
] <- mean(
  employees$Salary,
  na.rm = TRUE
)
```

Important: `na.rm = TRUE` tells R to ignore missing values while calculating the mean.

Method 3: Replace missing values with zero

```
employees$Salary[
  is.na(employees$Salary)
] <- 0
```

Note: Replace missing values with zero only when it makes business sense.

7. Removing Duplicate Records

Duplicate records are repeated rows in a dataset.

Check duplicates:

```
duplicated(employees)
```

Count duplicates:

```
sum(duplicated(employees))
```

Remove duplicates:

```
employees <- employees[
  !duplicated(employees),
```

```
]
```

8. Checking and Correcting Data Types

Sometimes imported data has incorrect data types.

Check structure:

```
str(employees)
```

Check a particular column:

```
class(employees$Salary)
```

Convert to numeric:

```
employees$Salary <- as.numeric(employees$Salary)
```

Convert to character:

```
employees$Name <- as.character(employees$Name)
```

Convert to factor:

```
employees$Department <- as.factor(
  employees$Department
)
```

9. Renaming Column Names

Use `names()` to change column names.

Display column names:

```
names(employees)
```

Rename a column:

```
names(employees)[
  names(employees) == "Name"
] <- "Employee_Name"
```

10. Export Cleaned Data

After cleaning, we can save the dataset as a CSV file.

Syntax:

```
write.csv(data, "filename.csv", row.names = FALSE)
```

Example:

```
write.csv(
  employees,
  "cleaned_employees.csv",
  row.names = FALSE
)
```

Check working directory:

```
getwd()
```

Change working directory:

```
setwd("D:/R Projects")
```

11. Practical Class Example

Create the following employee dataset:

```
employees <- data.frame(
  Employee_ID = c(101, 102, 103, 104, 105, 105),
  Name = c("Amit", "Rahul", "Priya", "Neha", "Vikas", "Vikas"),
  Department = c("Sales", "IT", "HR", "Sales", "IT", "IT"),
  Salary = c(30000, 45000, NA, 55000, 60000, 60000)
)
```

Task 1: Explore the dataset

```
head(employees)

str(employees)

summary(employees)

dim(employees)
```

Task 2: Find missing values

```
colSums(is.na(employees))
```

Task 3: Replace missing salary with average

```
employees$Salary[
  is.na(employees$Salary)
] <- mean(
  employees$Salary,
  na.rm = TRUE
)
```

Task 4: Remove duplicate records

```
employees <- employees[
  !duplicated(employees),
]
```

Task 5: Check cleaned data

```
head(employees)

str(employees)

summary(employees)

colSums(is.na(employees))
```

Task 6: Export cleaned data

```
write.csv(
  employees,
  "Final_Employee_Data.csv",
  row.names = FALSE
)
```

12. Student Practice Questions

1. Create an employee dataset with 10 records.
2. Import an employee CSV file into R.
3. Display the first 5 records.
4. Display the structure of the dataset.
5. Find the total number of missing values.
6. Find missing values column-wise.

7. Replace missing salary with average salary.
8. Remove duplicate records.
9. Convert the Salary column into numeric.
10. Rename the Name column to Employee_Name.
11. Export the cleaned dataset into CSV format.

13. Homework Assignment

Project: Employee Data Cleaning

Take any employee CSV or Excel dataset and complete the following:

- Import the dataset into R.
- Check rows and columns.
- Identify missing values.
- Remove duplicate records.
- Correct incorrect data types.
- Rename column names if required.
- Export the final cleaned dataset.

Expected Output: A cleaned employee dataset saved as `Final_Employee_Data.csv`.

14. Important Functions – Day 5

Function	Use
`read.csv()`	Import CSV
`read_excel()`	Import Excel
`head()`	First records
`tail()`	Last records
`str()`	Structure

`summary()`	Summary
`dim()`	Dimensions
`is.na()`	Check missing values
`sum()`	Count missing values
`colSums()`	Column-wise count
`na.omit()`	Remove missing rows
`duplicated()`	Identify duplicates
`as.numeric()`	Convert to numeric
`as.character()`	Convert to character
`write.csv()`	Export CSV
`getwd()`	Get working directory

`setwd()`	Set working directory

15. Data Analyst Workflow

RAW DATA → IMPORT → EXPLORE → CLEAN → VALIDATE → EXPORT

Remember: Before performing Data Analysis, always check the quality of your data.